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ECEN 5224
Week 15 Lab Report
04/22/26

Objective:

The objective of this lab was to analyze how channel insertion loss and impedance discontinuities (stubs) affect high-speed serial link performance, and to evaluate the effectiveness of transmit-side Feed-Forward Equalization (FFE). Simulations were performed in SerDes Designer using both a configurable S-parameter block and two provided S-parameter files representing a clean channel and a stub-loaded channel.

Method:

All simulations used SerDes Designer with NRZ modulation and differential signaling. Three metrics were captured for each configuration: the statistical eye diagram, the PRBS waveform, and the pulse response.

In the first part, the default S-parameter block was configured with a 10 dB insertion loss at the Nyquist frequency of 5 GHz. The pulse response, statistical eye diagram, and PRBS waveform were recorded. Insertion loss was then increased to 20 dB, which closes the eye, and a transmit FFE block was added to the simulation. Tap weights were manually tuned to compress the pulse response and open the eye as much as possible. Results before and after equalization were captured and compared.

In the second part, the analysis was repeated using two provided S-parameter files: K1_stub_no_15inch_FR408_bottom-routed-1.s4p, representing a 15-inch FR408 channel with no stub, and K2_stub_yes_15inch_FR408_stub-2.s4p, representing the same channel with a via stub present. For each file simulations were run without FFE and then with FFE applied. The symbol period was swept to determine the highest data rate at which a usable eye could be maintained.

Results:

Eye diagrams, pulse responses, and PRBS waveforms were captured for each configuration and are presented below:

10 dB Insertion Loss: Baseline

At 10 dB insertion loss, the channel introduces moderate low-pass filtering. The eye is open with visible ISI tails in the pulse response, but the statistical eye diagram shows clear separation and a well-defined eye opening.

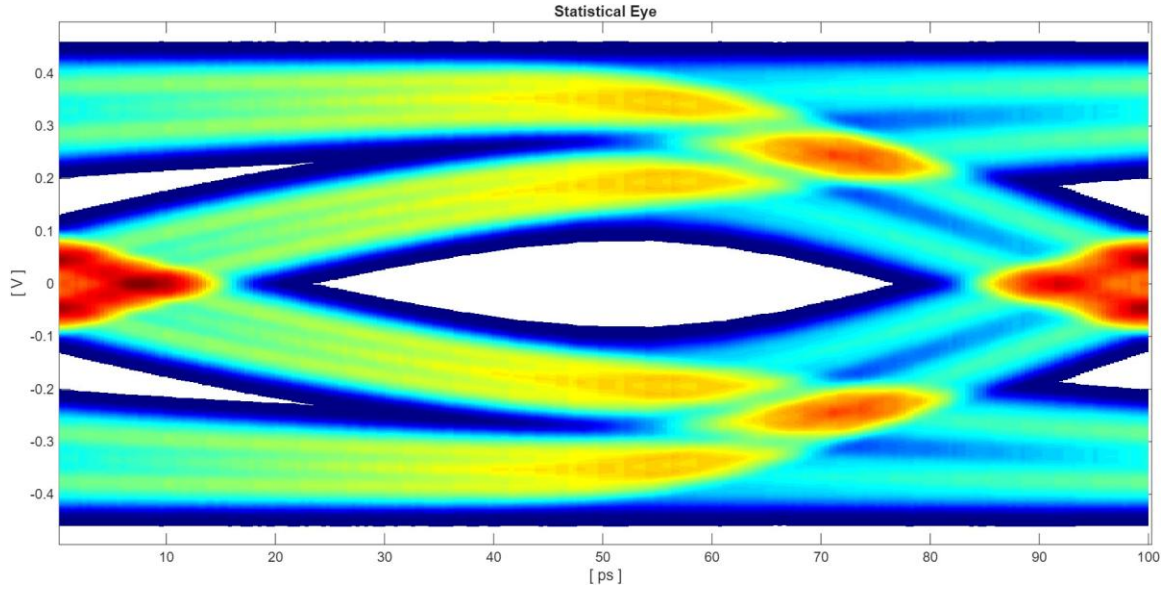


Figure 1: Statistical eye diagram at 10 dB insertion loss. Eye is open with a clear opening, indicating manageable ISI at this loss level.

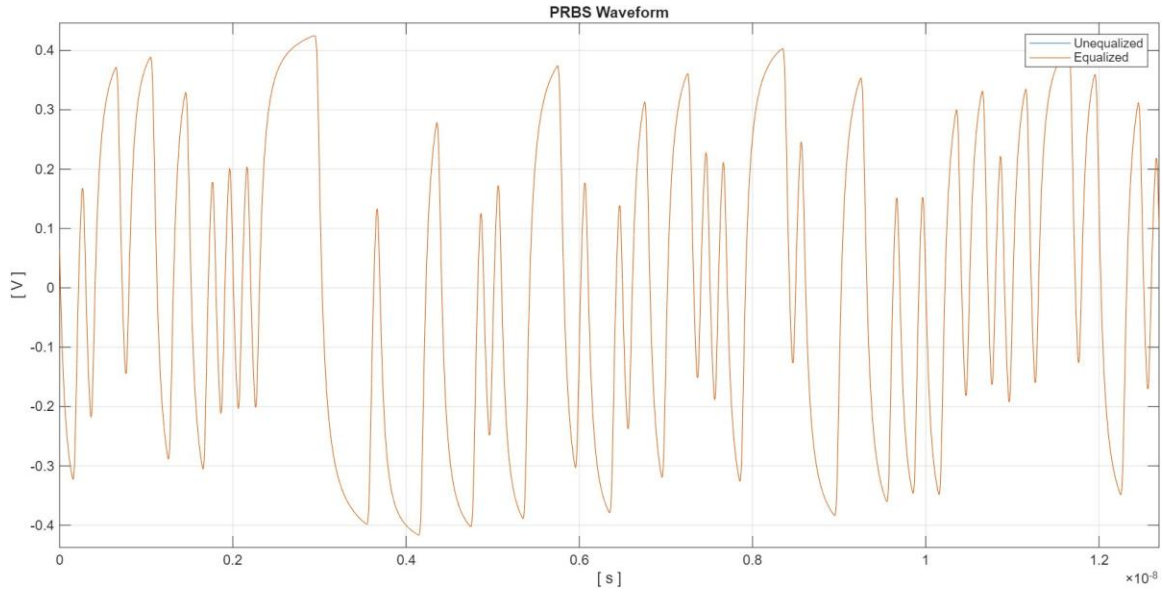


Figure 2: PRBS waveform at 10 dB insertion loss. The signal amplitude is reduced relative to the source, consistent with 10 dB of loss at Nyquist.

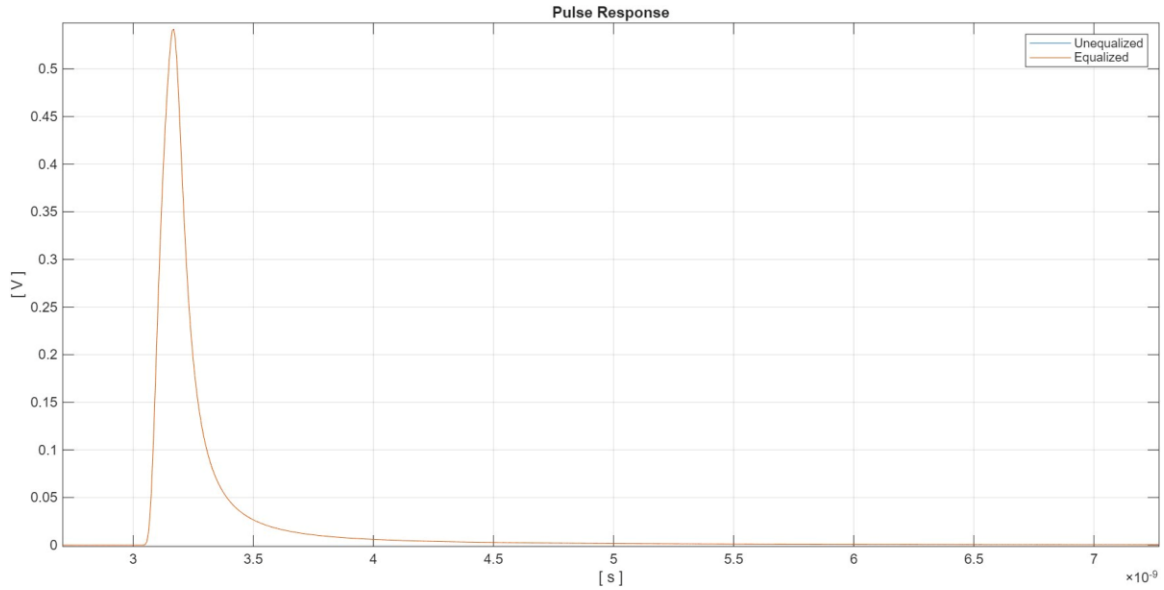


Figure 3: Pulse response at 10 dB insertion loss. The pulse decays with a moderate tail, indicating low-pass channel behavior with limited ISI spread.

20 dB Insertion Loss: Without FFE

Increasing the insertion loss to 20 dB substantially worsens the channel response. The pulse tail is significantly extended and the eye closes completely. The ISI from adjacent bits overwhelms the signal at the sampling point.

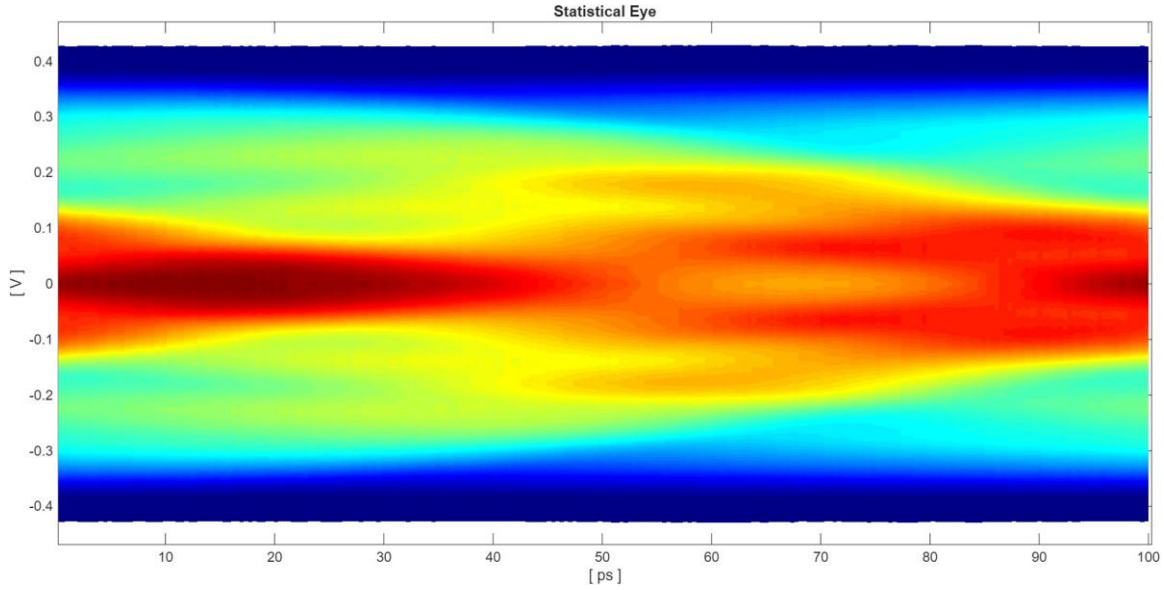


Figure 4: Statistical eye diagram at 20 dB insertion loss without FFE. The eye is fully closed; the transition density smears across the entire bit period.

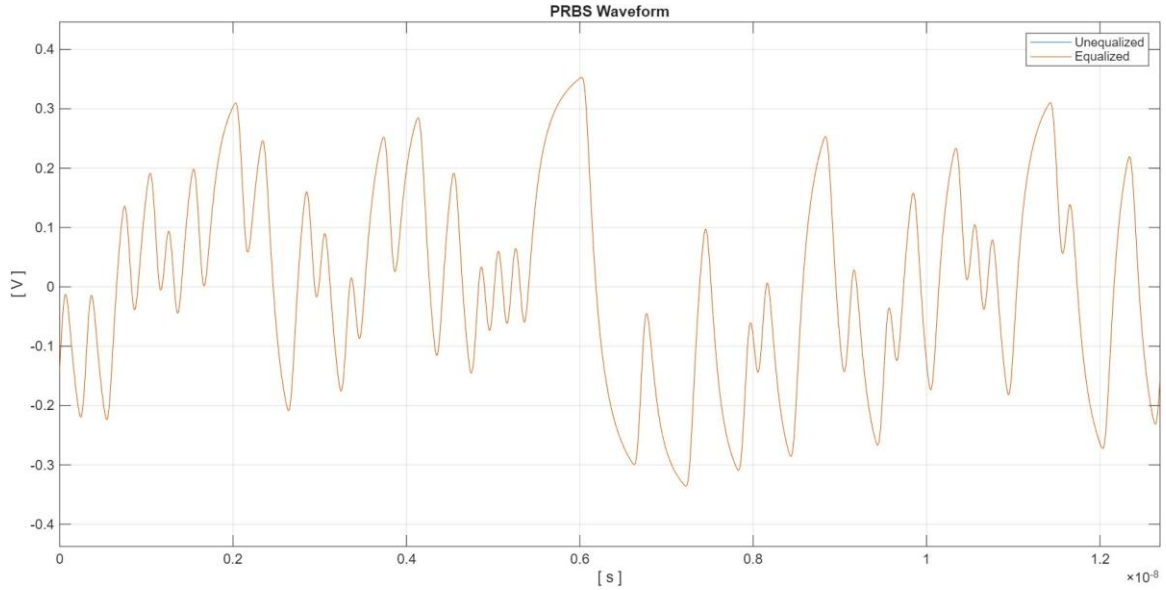


Figure 5: PRBS waveform at 20 dB insertion loss without FFE.

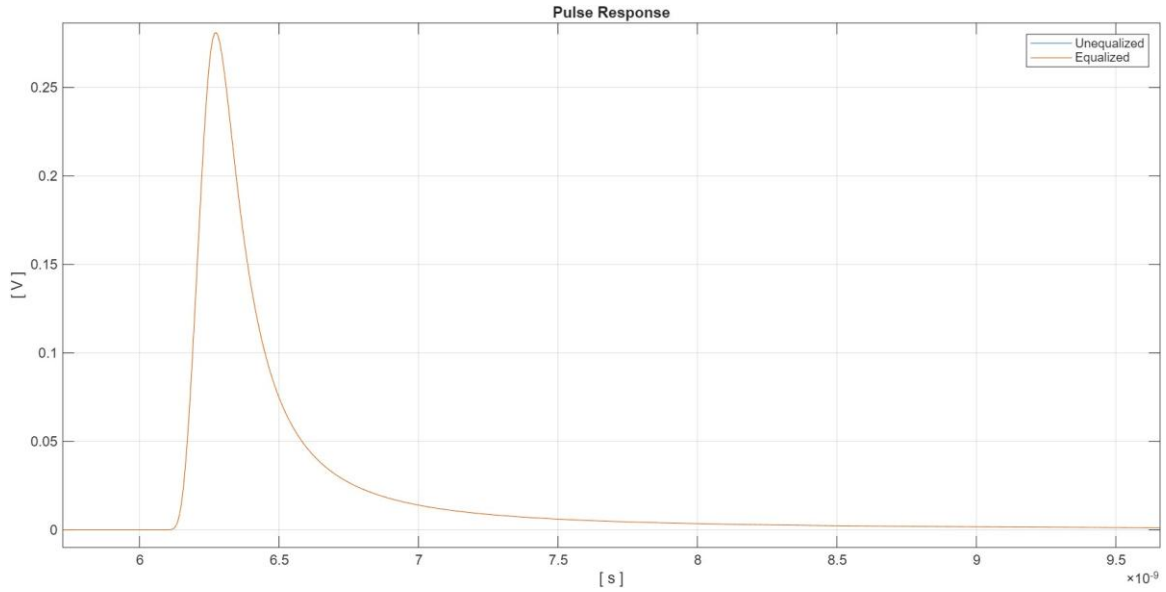


Figure 6: Pulse response at 20 dB insertion loss without FFE. The extended tail spans multiple bit periods, causing severe inter-symbol interference.

20 dB Insertion Loss: With FFE

An FFE block was inserted into the transmit path. Tap weights were manually tuned to pre-emphasize the transmitted waveform and compress the pulse tail. The final tap configuration used was [0, 1.075, −0.6, 0, 0] with T-spaced taps and tap normalization enabled, as shown in Figures 7 and 8.

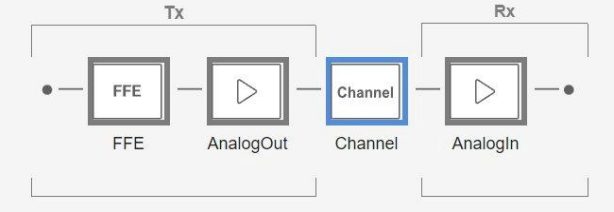


Figure 7: SerDes Designer block diagram showing the FFE block inserted in the transmit path, ahead of the analog output stage and channel.

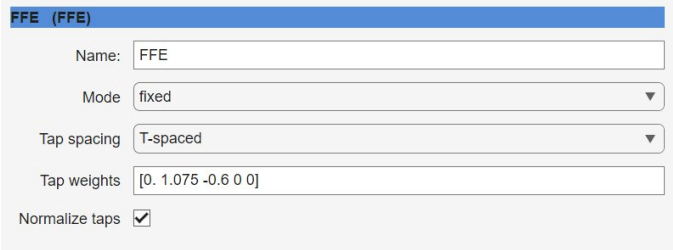


Figure 8: FFE configuration panel showing the manually tuned tap weights [0, 1.075, −0.6, 0, 0] with T-spaced taps and normalization enabled.

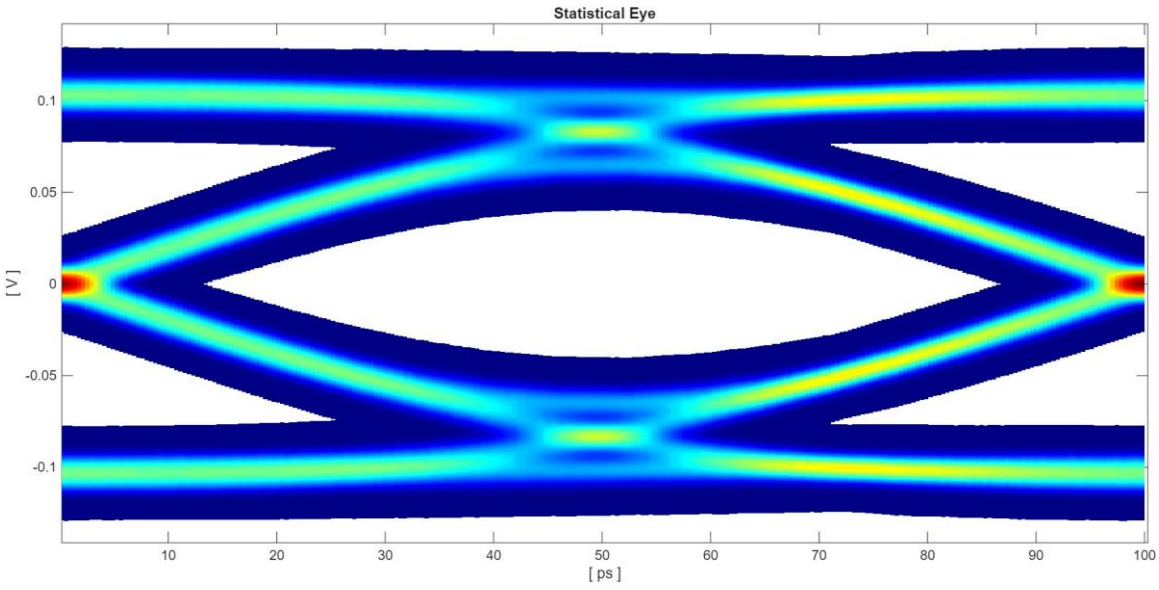


Figure 9: Statistical eye diagram at 20 dB insertion loss with FFE applied. The eye reopens clearly, demonstrating that FFE successfully mitigates ISI in a bandwidth-limited channel.

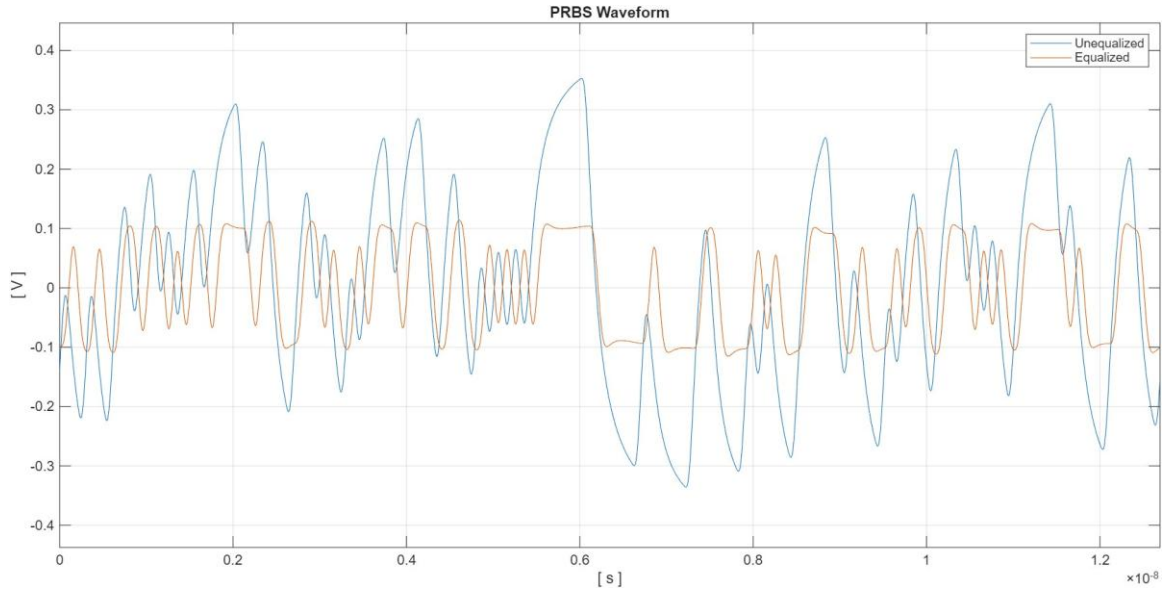


Figure 10: PRBS waveform at 20 dB insertion loss with FFE. The equalized signal (orange) shows reduced amplitude but improved transition sharpness compared to the unequalized signal (blue).

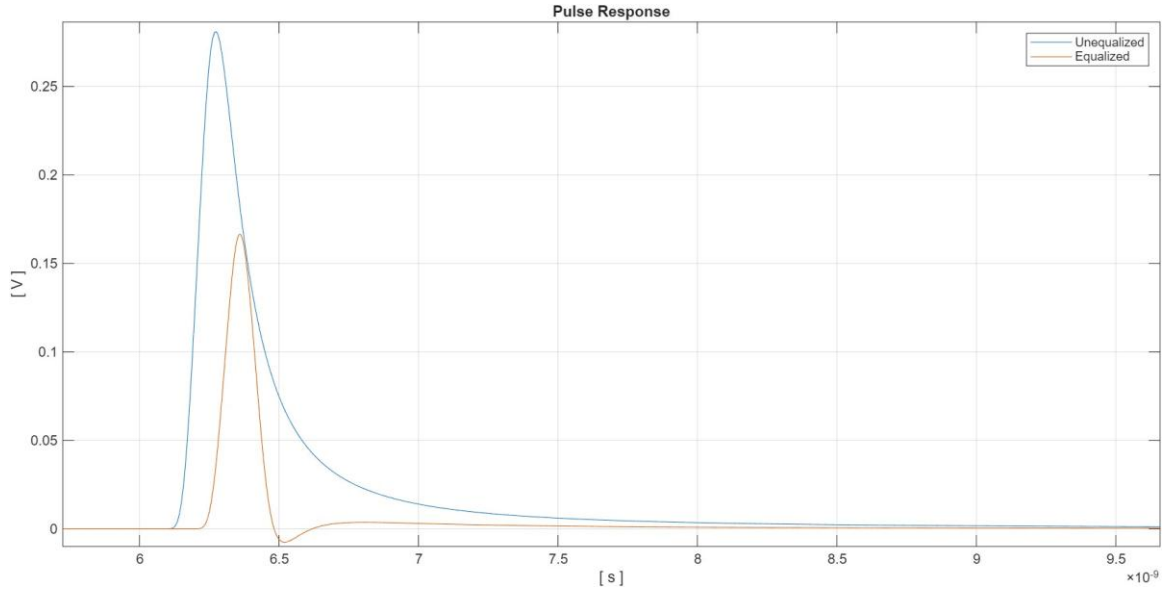


Figure 11: Pulse response at 20 dB insertion loss with FFE. The equalized pulse (orange) decays significantly faster than the unequalized channel (blue), confirming ISI suppression.

K1: No Stub Channel (15-inch FR408, Bottom-Routed)

The K1 S-parameter file represents a clean 15-inch FR408 channel with no stub. Without FFE the channel exhibits a long ISI tail. With FFE applied at a 60 ps symbol period (16.7 Gb/s), the pulse tail is compressed and the eye opens.

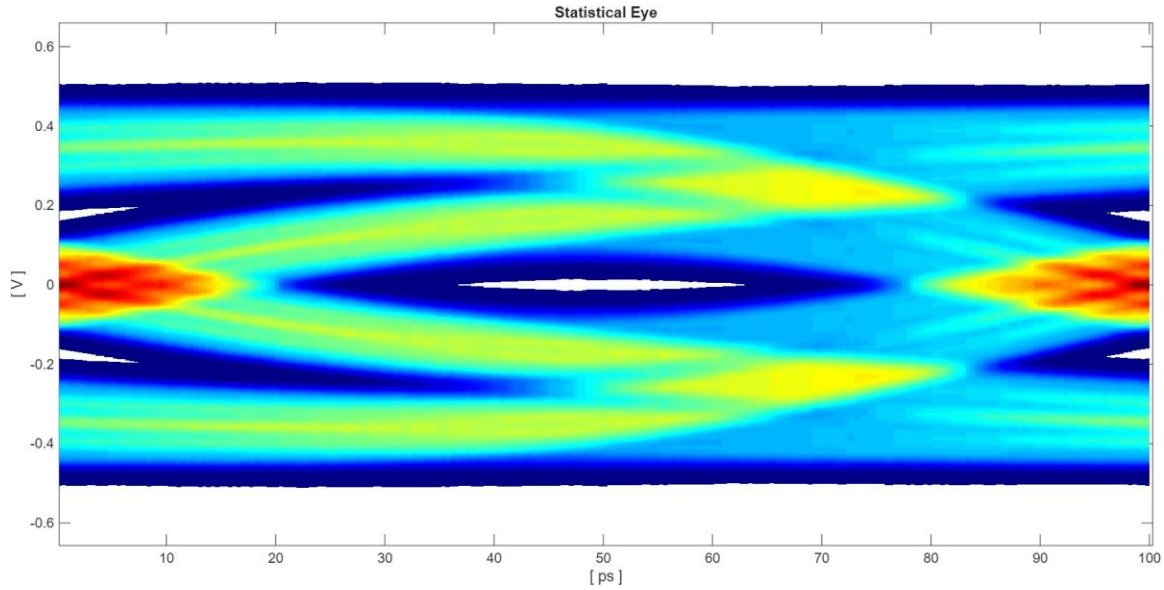


Figure 12: K1 statistical eye diagram without FFE. The eye is heavily degraded by ISI, with signal energy smeared across the bit period.

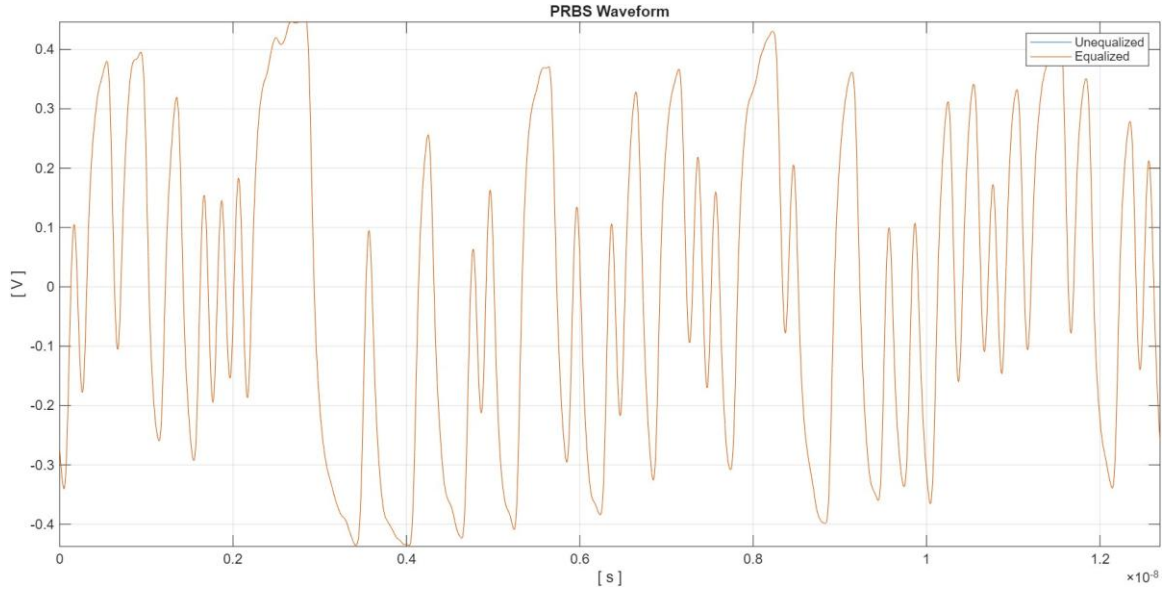


Figure 13: K1 PRBS waveform without FFE. Amplitude variation and ISI-induced rounding are visible across the bit stream.

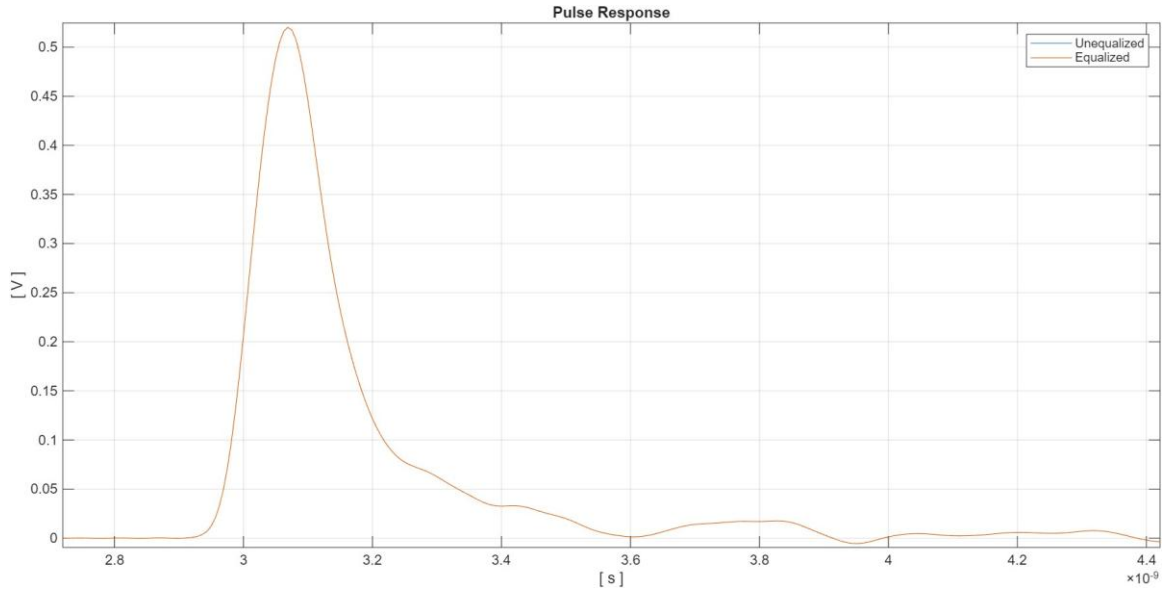


Figure 14: K1 pulse response without FFE. The pulse tail extends well beyond one bit period, causing inter-symbol interference at higher data rates.

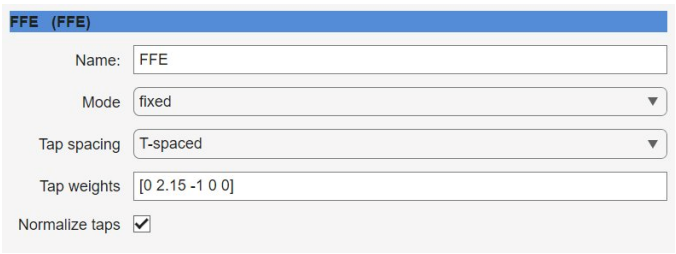


Figure 15: FFE tap configuration for K1 at 60 ps symbol period. Tap weights [0, 2.15, −1, 0, 0] with T-spaced taps and normalization enabled.

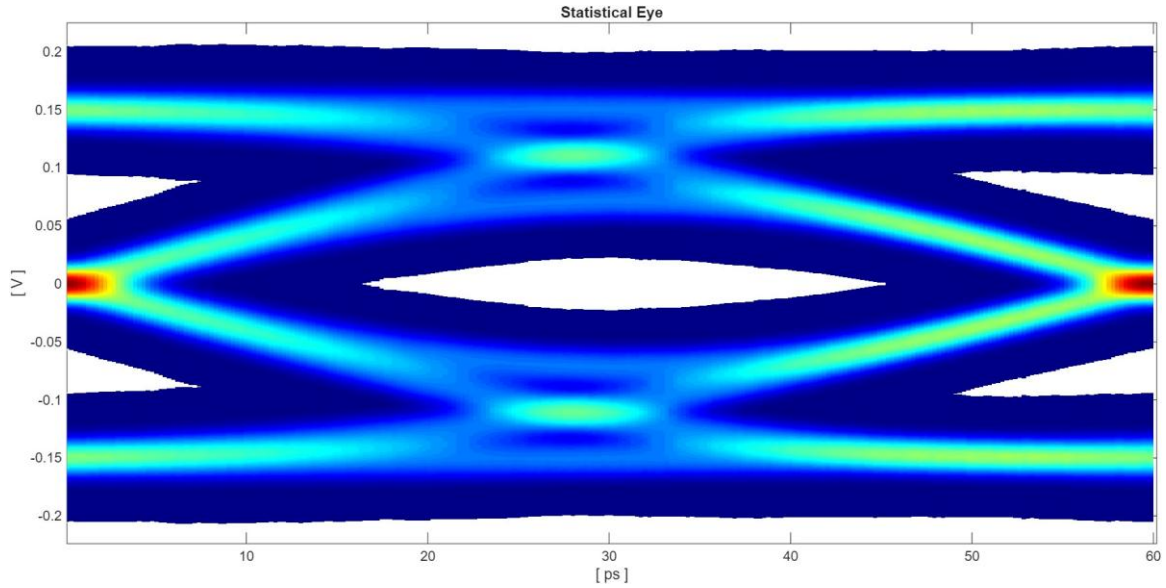


Figure 16: K1 statistical eye diagram with FFE at 60 ps (16.7 Gb/s). The eye opens clearly, demonstrating effective ISI cancellation in the stub-free channel.

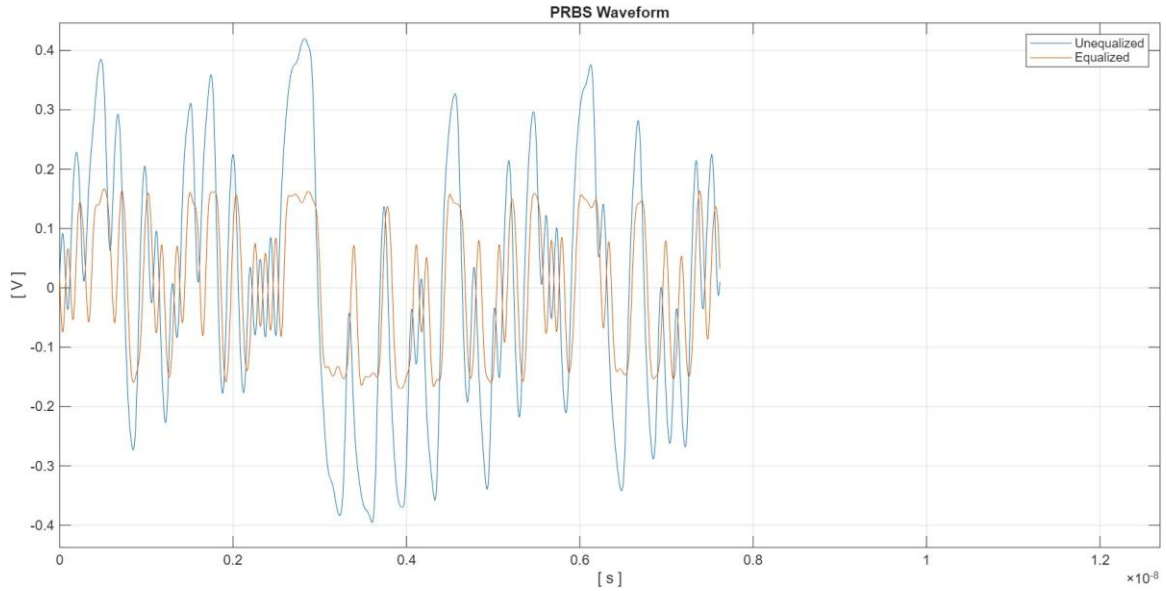


Figure 17: K1 PRBS waveform with FFE at 60 ps. The equalized output (orange) shows reduced swing but sharper transitions compared to the unequalized channel (blue).

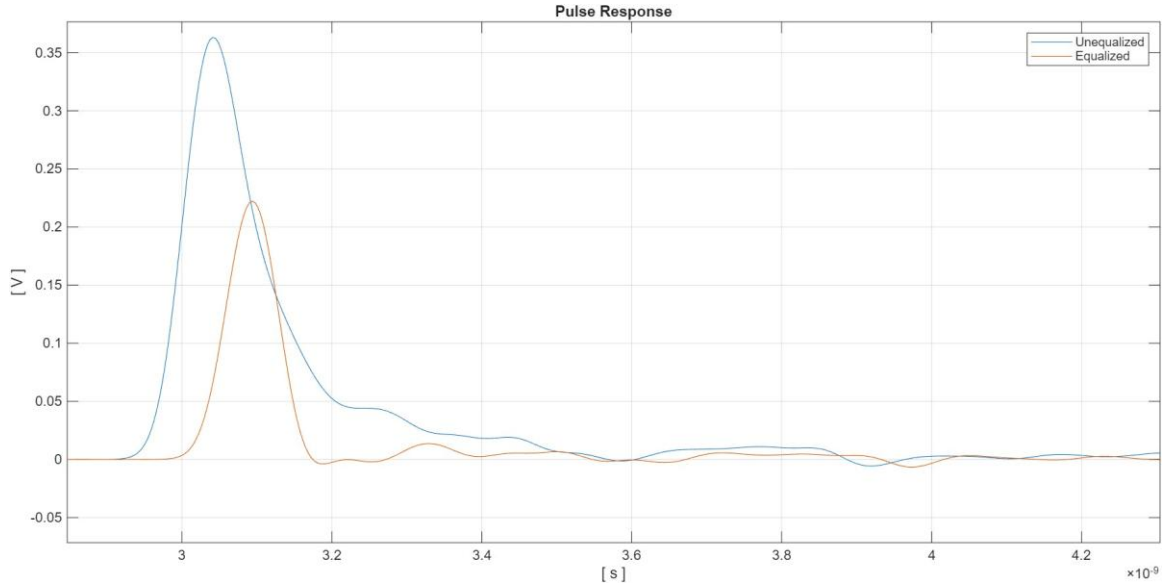


Figure 18: K1 pulse response with FFE at 60 ps. The equalized pulse (orange) decays rapidly to near zero within one symbol period, while the unequalized channel (blue) retains a persistent tail.

K2: Stub Channel (15-inch FR408, With Stub)

The K2 S-parameter file represents the same 15-inch FR408 channel but with a via stub present. Without FFE the stub introduces strong reflections that distort the signal and produce a closed eye. With FFE applied at a 130 ps symbol period (7.7 Gb/s), a marginal eye opening is achieved, but clear distortion from the stub remains visible.

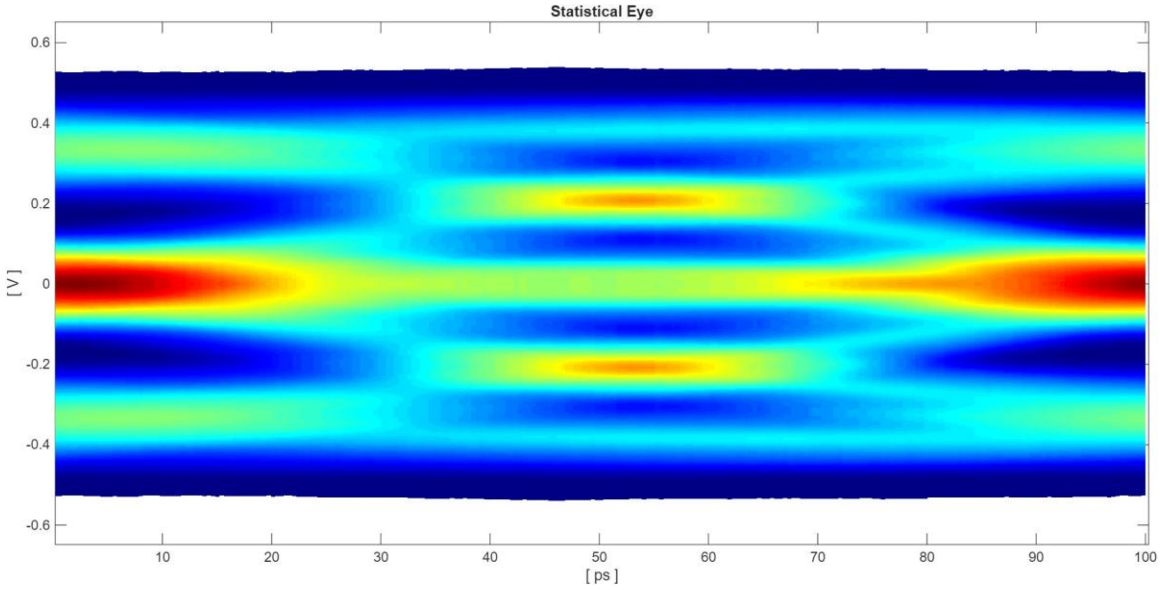


Figure 19: K2 statistical eye diagram without FFE. The eye is fully closed due to reflections from the stub. No usable eye opening exists at this symbol rate.

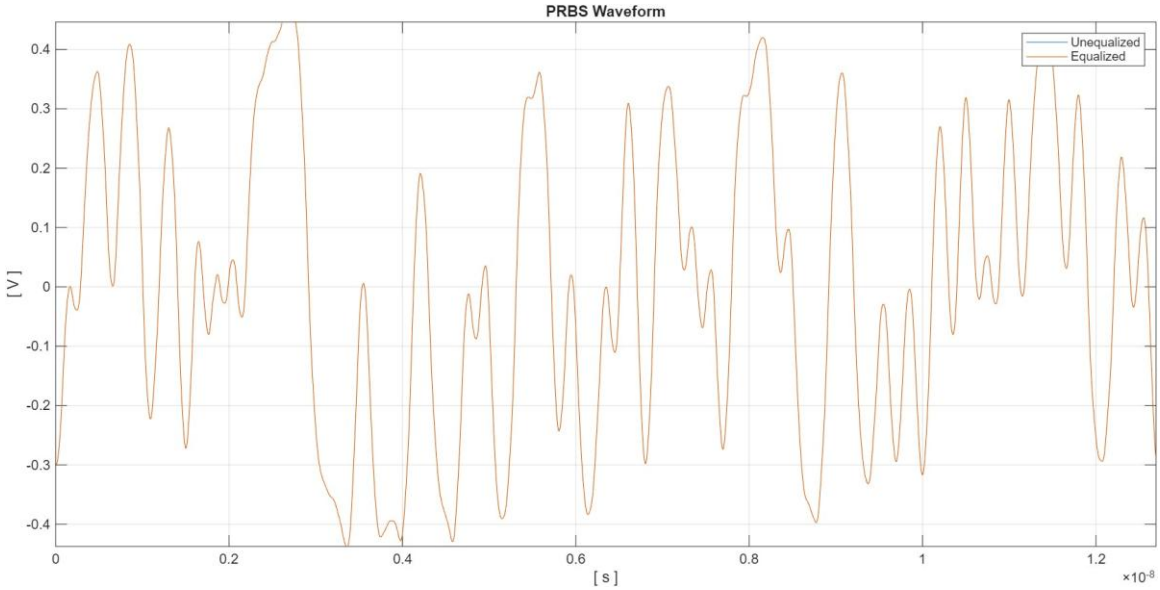


Figure 20: K2 PRBS waveform without FFE. The signal shows irregular amplitude variation and ringing consistent with impedance discontinuity reflections.

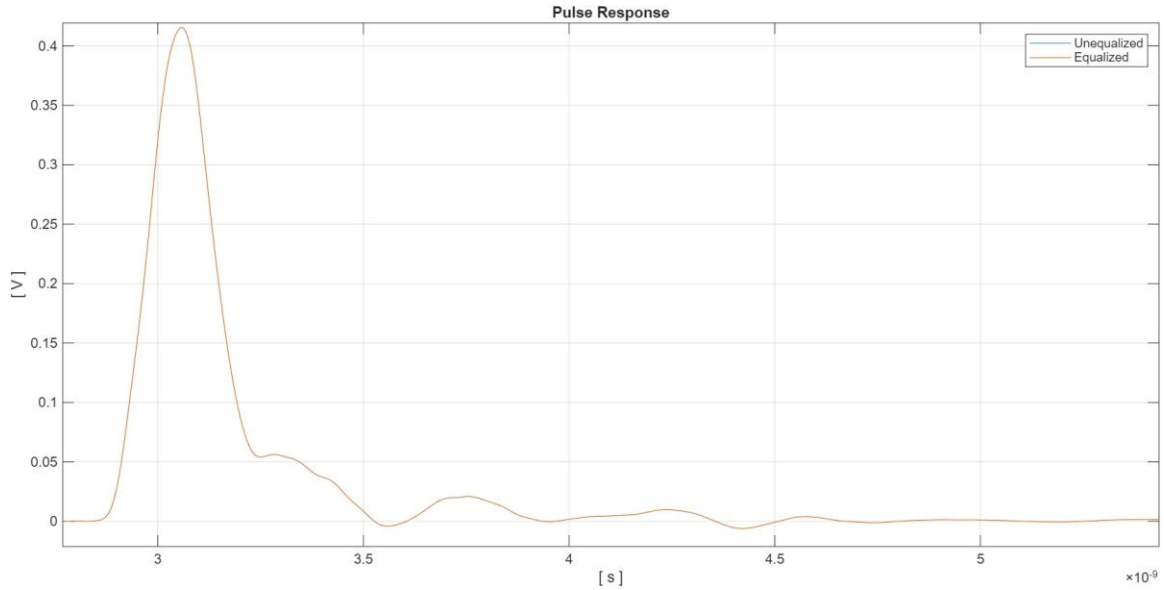


Figure 21: K2 pulse response without FFE. The extended tail with oscillatory behavior is characteristic of a stub-induced frequency null, where reflected energy re-enters the channel with a delay.

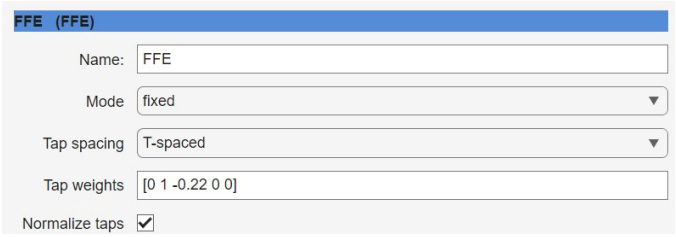


Figure 22: FFE tap configuration for K2 at 130 ps symbol period. Tap weights [0, 1, −0.22, 0, 0] with T-spaced taps and normalization enabled.

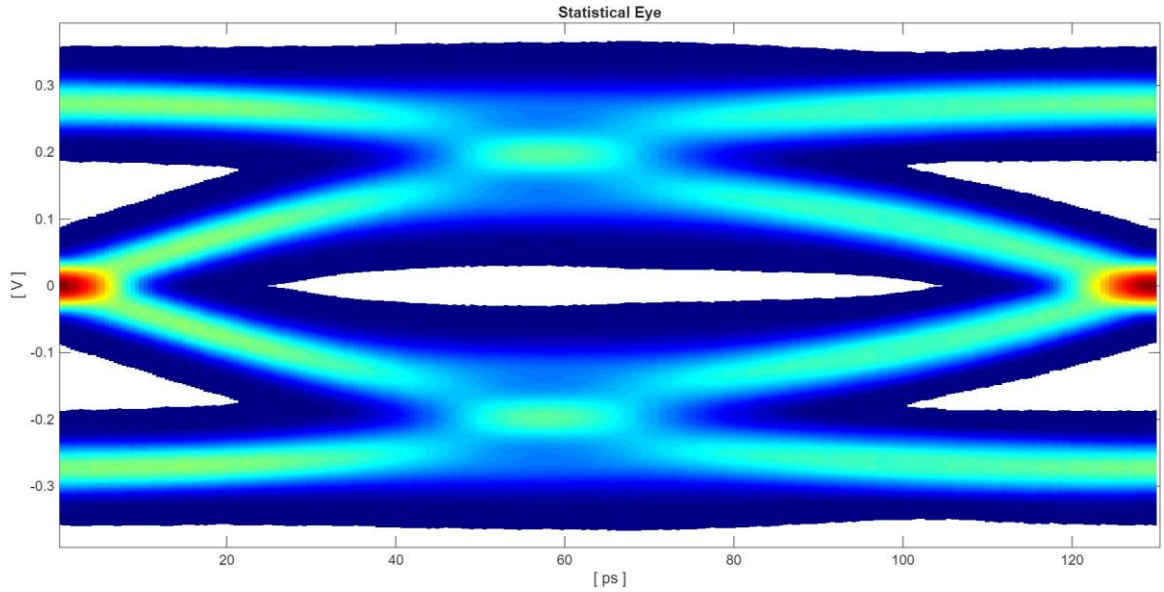


Figure 23: K2 statistical eye diagram with FFE at 130 ps (7.7 Gb/s). A narrow eye opening is achieved, but the eye is substantially smaller than the K1 result, reflecting the limits of FFE against stub-induced nulls.

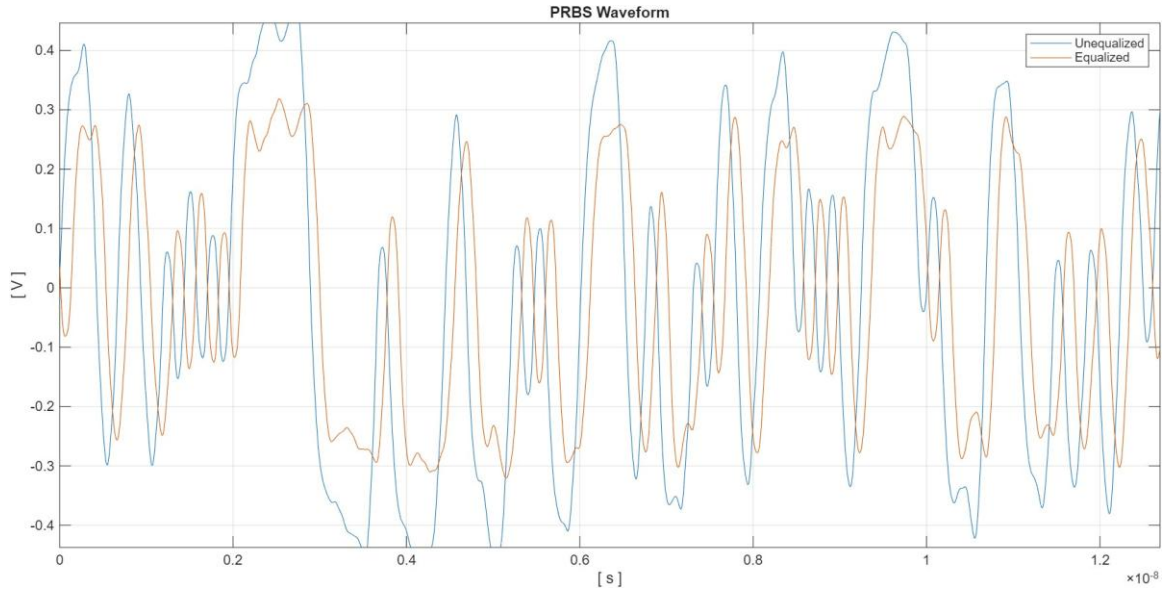


Figure 24: K2 PRBS waveform with FFE at 130 ps. The equalized signal (orange) shows reduced but still irregular amplitude, indicating residual distortion from the stub.

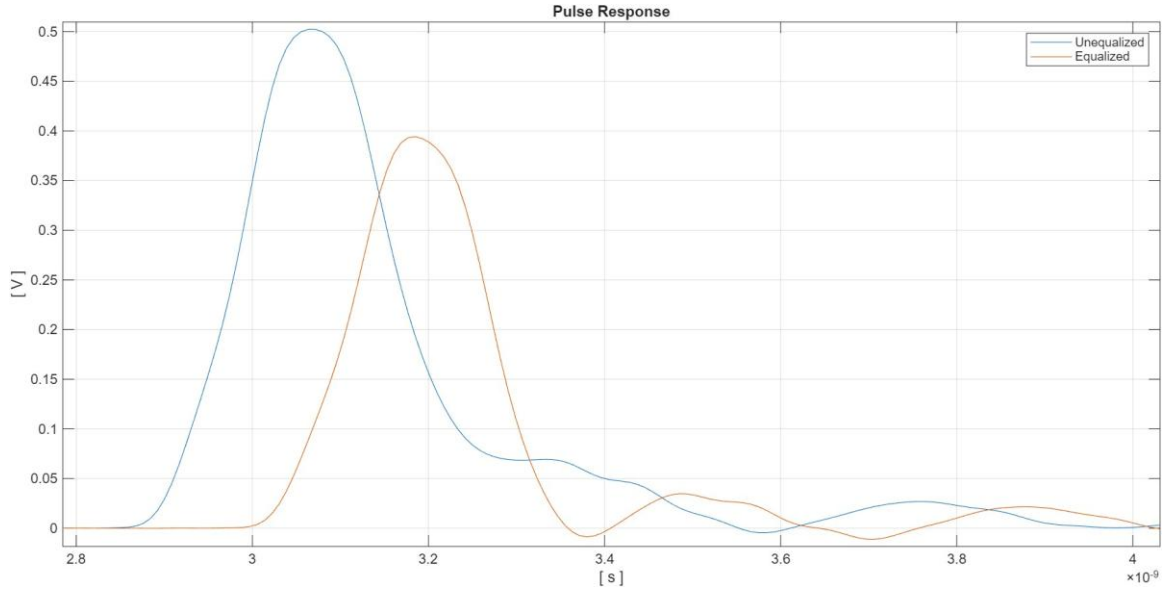


Figure 25: K2 pulse response with FFE at 130 ps. The equalized pulse (orange) is more compact than the unequalized channel (blue), but persistent low-level oscillations remain due to the spectral null introduced by the stub.

Table 1 summarizes the FFE tap weights used and the maximum achievable data rate for each S-parameter channel.

Channel	FFE Tap Weights [Pre, Main, Post1, Post2, Post3]	Max Data Rate (Gb/s)	Primary Limitation
K1 – No Stub	0, 2.15, -1, 0, 0	~16.7	Bandwidth / ISI
K2 – With Stub	0, 1.0, -0.22, 0, 0	~7.7	Stub reflections / nulls

Table 1: FFE tap weights and maximum achievable data rate for the K1 (no stub) and K2 (with stub) channels.

Analysis and Conclusion:

Increasing insertion loss from 10 dB to 20 dB substantially closed the statistical eye due to increased inter-symbol interference. FFE was able to counteract this by boosting the main tap and adding a negative post-cursor tap, the pulse response was compressed and the eye reopened. The pulse response comparison in Figure 11 shows this directly. The equalized trace decaying to near zero within one bit period while the unequalized channel tail remains for several bit periods. This confirms that ISI is correctable with transmit-side equalization.

The K1 S-parameter file, representing the clean 15-inch FR408 channel, responded well to FFE. The pulse tail was shortened, and a maximum data rate of approximately 16.7 Gb/s was achievable, though a more conservative operating range of 12 to 14 Gb/s would be best.

The K2 S-parameter file, which includes a stub, produced a different result. The stub acts as an impedance discontinuity that reflects energy which creates nulls in the channel's frequency response. These nulls represent signal energy that is permanently lost rather than delayed. FFE operates by shaping the transmitted waveform in time but can't recover information the channel response is zero. This means that even with FFE applied, the maximum achievable data rate for the stub channel is only around 7.7 Gb/s, less than half of the clean channel.

These simulations show that a stub-loaded channel on the same board with the same trace length supports less than half the data rate of a clean-routed channel, regardless of how equalization is applied.